Alice Perego

Ph.D. Student in Astrophysics
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Main Scientific Interests

LISA mission - ultra-compact double white dwarfs - multi-messenger astrophysics - photometric and spectroscopic surveys - LISA data analysis - stellar binary evolution - Milky Way - gravitational wave backgrounds - massive black hole binaries - extreme mass-ratio inspirals

Education

Université Côte d'Azur Nice, France, November 2024 – present

Ph.D. in Astrophysics – Expected Graduation: Autumn 2027

- Thesis title: Future perspectives on gravitational-wave astrophysics with space-based detectors: enabling multi-messenger studies of Galactic binaries
- Supervisors: Nelson Christensen, Astrid Lamberts

University of Milano Bicocca Milan, Italy, October 2022 – October 2024

Master's Degree in Astrophysics and Space Physics – Final grade : 110/110 cum laude

- Thesis title: Assessing the performance of future space-based detectors: astrophysical foregrounds and individual sources
- Supervisors: Alberto Sesana, Matteo Bonetti, Sylvain Marsat

University of Milano Bicocca Milan, Italy, October 2019 – September 2022

Bachelor's Degree in Physics – Final grade : 110/110 cum laude

- Thesis title: Numerical simulations of oscillating neutron stars
- Supervisor: Bruno Giacomazzo

Research Experience

Observatoire de la Côte d'Azur Nice, France, November 2024 – present

Doctoral Research

- My work focuses on bridging LISA detections of white dwarf binaries with electromagnetic (EM) observations: I developed a methodology to identify LISA candidates in multiband photometric surveys. At present, I am working on simulated LISA and EM catalogues to optimise the cross-matching of multi-messenger systems. My work actively contributes to observational strategies, including an ESO proposal for P118 and I am designing a large-scale survey with the Canada France Hawaii Telescope. In addition, I am actively contributing to the scientific evaluation of proposed mission designs for future space-based GW detectors within ESA Voyage 2050 project and I am part of the GUEST proposal for ESA 2025 Call for Missions (F-class)

Laboratoire des 2 infinis (L2IT) Toulouse, France, March 2024 – July 2024

Research internship

– My work involved modelling gravitational-wave background signals from micro-Hertz to deci-Hertz frequencies. I developed a unified framework to perform a self-consistent computation of these signals generated by a broad set of populations, such as massive black hole binaries, extreme mass-ratio inspirals and stellar-mass binaries. In th context of the ESA project Voyage 2050, I assessed their impact on future space-based GW detectors and performed parameter estimation of massive black holes binaries.

Observational Proposals

- Canada-France-Hawaii Telescope Community Survey (2027B-2033A)
  Work Package on *Stellar and Binary Evolution in the Milky Way*
  PI: Alice Perego **approved by SAC**
  *CFHT/Megacam – **172 high-quality nights***
- Binary white dwarfs: from single systems to a LISA population (ESO P118)
  PI: Alice Perego allocation pending
  ESO/UVES – 152 hours

Conferences, Workshop & Research Visits

- The VI Gravi Gamma Nu workshop Milan, Italy – October 2026
  Contributed talk
- European Astronomical Society Annual Meeting Lausanne, Switzerland – July 2026
  Contributed talk
- LISA Symposium College Park, USA – June 2026
  Contributed talk
- LISA France Paris, France – May 2026
  Contributed talk
- LISA Astrophysics Working Group meeting Geneva, Switzerland – December 2024
  Contributed talk
- GW Space 2050 workshop Potsdam, Germany – November 2025
  Oral presentation
- "Enabling Future Gravitational Wave Astrophysics in the Milli-Hertz Regime" workshop Garching, Germany – July 2025
  Oral presentation
- LISA Sprint workshop Huntsville, USA – April 2025
- LISA Distributed Data Processing Center (DDPC) workshop Milan, Italy – June 2025
- Research visit at Carnegie Mellon University Pittsburgh, USA – April 2025
  Host: Katelyn Breivik
- Annual meeting of the French Society of Astronomy & Astrophysics Toulouse, France – July 2025
  Contributed talk – published proceeding: Preparing for LISA with CFH telescope: Synergies between gravitational waves and electromagnetic observations
- Rencontres de Moriond - Gravitation La Thuile, Italy – April 2025

Contributed talk – published proceeding: Assessing the performance of future space-based detectors: astrophysical foregrounds and individual sources

- LISA Astrophysics Working Group meeting — Garching, Germany – November 2024
  Contributed talk
- GWOSC Gravitational Wave Open Data workshop — Toulouse, France – April 2024
  Mentor at an in-person study hub

Seasonal Schools & specialised training

- LISA School for early-career scientists — Les Houches, France – October 2025
- MOOC "Reproducible Research II: Practices and tools for managing computations and data" — Online – September 2026
- MaNiTou Summer School on Gravitational Waves — Toulouse, France – July 2024
- Masterclass on Big Data within Science and Industry — Milan, Italy – September 2023

Membership

- LISA Consortium - core member
- LISA DDPC - L3C coordination unit
- LISA Astrophysics working group
- GW Space 2050 working group, within ESA *Voyage2050* project
- GUEST proposal team, within ESA 2025 Call for Missions (F-class)

Software Skills

- **Programming:** Python, C++
- **Softwares and codes:** Standard scientific Python packages (astrophysics, data analysis, machine learning), COSMIC, LISABETA, GBGPU, EXTRAPOPS (public codes for population synthesis and GW data analysis), PHOTOCAT and GWPOP_TOOLS (proprietary pipelines for WD modelling and GWB and SNR calculations), VizieR and Aladin (astronomical dataset and visualisation tools)
- **Analysis techniques:** Frequentist and Bayesian statistical inference

Languages

- **Italian:** Native
- **English:** Fluent
  - Certification: Bbetween English C1 Open Badge awarded by the University of Milano Bicocca
- **French:** Intermediate
  - Certification: French proficiency test B2 awarded by the GoFluent online course

Publications, Research Outputs and Proceedings

Perego, A., Bonetti, M., Sesana, A., Toonen, S. & Korol, V. Assessing the performance of future space-based detectors: Astrophysical foregrounds and individual sources. *Physical Review D* **114,** 023022. doi:10.1103/nvgj-8pmc. arXiv: 2510.18695 [astro-ph.HE] (July 2026).

Perego, A., Lamberts, A., Schultheis, M. & Christensen, N. Towards systematic searches for LISA white dwarf binaries with multiband photometry. *Astronomy & Astrophysics* **701,** L6. doi:10.1051/0004-6361/202556494. arXiv: 2509.09348 [astro-ph.HE] (Sept. 2025).

Perego, A. *et al.* Crossmatching LISA detections of white dwarf binaries with electromagnetic catalogues. *in prep.*

Blas, D. *et al.* GUEST: Gravitational Universe Exploration with Satellite Tracking. A passive satellite laser-ranging mission for the dark gravitational Universe. *arXiv e-prints,* arXiv:2607.18390. doi:10.48550/arXiv.2607.18390. arXiv: 2607.18390 [astro-ph.CO] (July 2026).

Coughlin, S. *et al. COSMIC-PopSynth/COSMIC: v3.6.1* version v3.6.1. July 2025. doi:10.5281/zenodo.15855958.

Perego, A., Lamberts, A., Schultheis, M. & Christensen, N. Preparing for LISA with CFH telescope: Synergies between gravitational waves and electromagnetic observations. In *SF2A-2025: Semaine de l'Astrophysique Francaise* (eds Siebert, A. *et al.*) 421–424 (Dec. 2025).

Perego, A. Assessing the performance of future space-based detectors: astrophysical foregrounds and individual sources. In *59th Rencontres de Moriond on Gravitation: Moriond 2025 Gravitation* (eds Augé, E., Dumarchez, J. & Trân Thanh Vân, J.) 119–122 (June 2025).